

Math 142, S14

Quiz 6

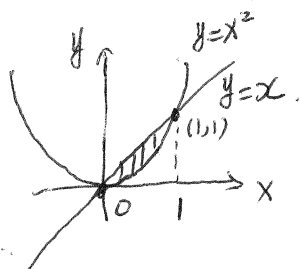
Section:

Name: Answer

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Sketch the region enclosed by $y = x$, $y = x^2$ and find its area.

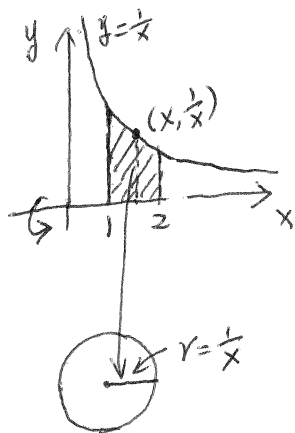


$$\begin{aligned} \begin{cases} y=x \\ y=x^2 \end{cases} &\Rightarrow x=x^2 \Rightarrow x^2-x=0 \\ &\Rightarrow x(x-1)=0 \\ &\Rightarrow x=0 \text{ or } 1 \end{aligned}$$

Thus

$$\begin{aligned} A &= \int_0^1 x - x^2 dx \\ &= \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_{x=0}^{x=1} \\ &= \left(\frac{1}{2} - \frac{1}{3} \right) - 0 = \frac{1}{6} \end{aligned}$$

Problem 2. (5 points) Find the volume of the solid obtained by rotating the region bounded by $y = 1/x$, $y = 0$, $x = 1$, $x = 2$ around the x -axis.



$$\begin{aligned} A(x) &= \pi \left(\frac{1}{x} \right)^2 = \frac{\pi}{x^2} \\ \Rightarrow V &= \int_1^2 A(x) dx \\ &= \int_1^2 \frac{\pi}{x^2} dx \\ &= \pi \left[-\frac{1}{x} \right]_{x=1}^{x=2} \\ &= \pi \left[-\frac{1}{2} + 1 \right] \\ &= \frac{\pi}{2} \end{aligned}$$